

Bálits Péter Márton

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EDUCATION

Eötvös Loránd University

Bachelor of Arts in Computer Science

Budapest, HU

Sep. 2021 – July 2026

- Thesis: wrote a 40K lines-of-code etcd clone in Go from scratch, using HashiCorp'sraft library and Kubernetes

Czech Technical University

Erasmus semester at the Faculty of Informatics

Prague, CZ

Sep. 2024 – Febr. 2025

- Built a CSV spreadsheet engine in Scala focusing on expression parsing, evaluating and cell arithmetics

EXPERIENCE

Backend Developer

Fornax ICT

Oct. 2025 – July. 2026

Budapest, HU

- Developed and maintained backend services using Java, Spring Boot and PostgreSQL.
- Contributed to internal projects with mature codebases, designed APIs, optimized query performances.

PROJECTS

Kave | Distrubted key-value store with MVCC, Leases, Watches and Oblivious Transfer
Go, Svelte

Jan. 2026 – May 2026

- * Distributed Raft cluster over Kubernetes with failure tolerance, rolling updates and pod distribution
- * MVCC as the fundamental concurrency primitive, used for global monotonic revisions, point-in-time reads and transactions
- * Leases for attaching TTLs to keys
- * Watches for streaming real time changes to subscribed keys over WebSocket
- * Oblivious Transfer backed secret storage
- * HTTP API endpoints, with follower-to-leader proxying, built-in Prometheus metrics,
- * docker-compose for local development, stress and smoke tests, GitHub Actions for CI/CD

Logregator | Time-series log aggregator over TCP

Rust, Tokio, Glommio

May 2026 – Aug. 2026

- * High throughput async TCP log aggregator server designed append-heavy workloads
- * Custom time-series LSM-tree storage engine, including SSTables for persistence and WAL for crash recovery
- * Label indexed logs inspired by Grafana Loki for fast queries
- * Custom benchmarking suite for fine-grain performance tuning and visualizing
- * Used cargo workspaces with subcrates for storage engine, tcp server/client, internal benchmarking

TECHNICAL SKILLS

Languages: Rust, Go, Scala, Java, Python, C++, SQL, TypeScript

Technologies: Linux, git, make, cargo, Docker, Kubernetes, GitHub Actions, (?perf, strace, pprof?)

Systems, protocols: Raft consensus, TCP/IP, HTTP, (?gRPC?) distributed storage, write-ahead logs

Spoken languages: Hungarian (native), English (fluent), German (B2)